

11	<i>either</i> quartz <i>or</i> piezo-electric crystal	B1	
	opposite faces /two sides coated (with silver) to act as electrodes	B1	
	<i>either</i> molecular structure indicated		
	<i>or</i> centres of (+) and (–) charge not coincident	B1	
	potential difference across crystal causes crystal to change shape	B1	
	alternating voltage (in US frequency range) applied across crystal	B1	
	causes crystal to oscillate / vibrate	B1	
	(crystal cut) so that it vibrates at resonant frequency	B1	[6]
	(max 6)		

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12 (a) signal becomes distorted / noisy

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